

Whenever I open Instagram on my smartphone, several network devices work together to ensure that photos, videos, reels, and messages load quickly and securely.

1. Access Point (Wi-Fi)

If I am connected to my home Wi-Fi, my phone first communicates with the Access Point (AP), which is usually built into the Wi-Fi router.

The access point allows my phone to connect wirelessly to the local network using Wi-Fi.

Without the access point, my phone would not be able to access the home network wirelessly.

2. Router

After receiving data from the access point, the Router sends my Instagram request from my home network to my Internet Service Provider (ISP) and then to Instagram's servers.

When Instagram sends back photos, videos, or messages, the router receives the data and forwards it to my phone.

The router also assigns my device an IP address (using DHCP) and allows multiple devices in my home to share one Internet connection (using NAT).

3. Firewall

A Firewall, often built into the router, checks incoming and outgoing network traffic.

It blocks unauthorized access and suspicious traffic while allowing safe Instagram communication.

This helps protect my phone and home network from cyber attacks.

4. Switch

Although I don't usually see it at home, Switches are heavily used inside my ISP's network and Instagram's data centers.

Switches efficiently forward data between servers and networking devices using MAC addresses.

They help Instagram deliver images, videos, and messages quickly by reducing unnecessary network traffic.

